

Dash - Introduction and Installation

Install dash and pandas

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/dash$ pip install dash
Collecting dash
  Downloading dash-3.1.1-py3-none-any.whl (7.9 MB)
    | 7.9 MB 16.5 MB/s
Collecting plotly>=5.0.0
  Downloading plotly-6.2.0-py3-none-any.whl (9.6 MB)
    | 9.6 MB 26.0 MB/s
Collecting Flask<3.2,>=1.0.4
  Downloading flask-3.0.3-py3-none-any.whl (101 kB)
    | 101 kB 7.0 MB/s
Requirement already satisfied: typing-extensions>=4.1.1 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from dash) (4.13.2)
Requirement already satisfied: importlib-metadata in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from dash) (8.5.0)
Requirement already satisfied: requests in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from dash) (2.32.3)
Requirement already satisfied: setuptools in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from dash) (44.0.0)
Collecting Werkzeug<3.2
  Downloading werkzeug-3.0.6-py3-none-any.whl (227 kB)
    | 227 kB 31.1 MB/s
Requirement already satisfied: nest-asyncio in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from dash) (1.6.0)
Collecting retrying
  Downloading retrying-1.4.0-py3-none-any.whl (11 kB)

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/dash$ pip install pandas
Requirement already satisfied: pandas in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (2.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: numpy>=1.20.3; python_version < "3.10" in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from pandas) (1.24.4)
Requirement already satisfied: pytz>=2020.1 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.1 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

Create file my_first_dash_app.py

Launch and view the app

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/dash$ python3 my_first_dash_app.py
Dash is running on http://0.0.0.0:5000/

* Serving Flask app 'my_first_dash_app'
* Debug mode: on
```

Open URL <http://localhost:5000/> to view 'First API Dash' page

First API Dash